Manual

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Contents

1 Flatcam:

1.1 calculate tool diameter

tool→calculator input tip-Diameter as 0.2mm input tip angel as 60 input desired cut-z as 0.1 use the tool diameter given in isolating routing

1.2 Isolation Routing

File→Open→Open Gerber in Project tab select the gerber file that has your isolation routing select isolation routing under Tools in gerber object isolation In isolation tool tool add from DB and input the tool Dia that was calculated in the tool calculator

press the generate geometry button

in parameters for Tool 1, select the dia of the v-tip and input 0,02 then input the angle of the v-tip 60, both both x-y and z feed rates between 0-170 (170 recommended) set the spindle speed at 11000. press Generate CNCJob object save the cnc code for the usb drive.

1.3 Drilling

File→Open→Open Excellon select drilling Tool

select the tools you want to create the CNC job fore

Adjust Drill Z to the thickness of the copper laminate(value must be negative)

set travel Z to 2.00 mm

set feed rate between 0 and 170(170 recommended) set spindle speed to 11000.

click Create CNC Job object

save the cnc code for the usb drive

1.4 Board Cutout

File→→OpenOpen Gerber for your board cutout select your board outline Under Tools select cutout tool specify a Margin. This will create a rectangular cutout at the given distance from any element in the Gerber. Specify a Gap Size. 2 times the diameter of the tool you will use for cutting is a good size. Specify how many and where you want the Gaps along the edge, 2 (top and bottom), 2 (left and right) or 4, one on each side.

Click on Generate Geometry.

Create a CNC job for the newly created geometry

1.5 2-side PCB

open File→Open→Open Gerber Select the side that has not bin used (copper top/bottom)

select Tools→2-sided PCB tool under type select gerber and the side of the pcb you are working on Then at reference select box and under reference object select gerber and board outline (depends on file name) and then select mirror.

Select drill dia (usually 1 or 1.5mm). Select alignment drill coordinates by right click att the coordinate than press +aff Generate excellon object.

Go to Project select the gerber file that was mirrored proceed as in isolation routing

Go to Project select the alignment hole file(name my differ) proceed as in drilling

2 Wegster operation:

2.1 start up

load the g-code files that you wish to use onto the computer in a way that you can find them. Open Wegstr CNC controlling software if it is not already open. if the indicator under status does not read device ready push the black button next to the power button on the wegstr CNCMilling Machine clamp down the copper laminate to the machine bed locate the bits with corresponding dimensions as the tools used in FlatCAM

2.2 safety

The manufacturer recommends the use of eye protection and protective gloves while operating the machine.

2.3 Isolation Routing

change the bit to the spital bit load the g-code for your isolation routing push the button in the software under the text auto level than press the button labeled start use the vacuum to remove the fiberglass recommended to use while the Routing is tacking place

2.4 Drilling

load the g-code for the drilling

change to the drill bit

if the isolation routing has bin used, use the x and y movement controls to move the spindle to x = 0 and y = 0 and than use the z axis control to move the drill bit down to the copper and set z = 0

else push the auto level button

than press the button labeled start use the vacuum to remove the fiberglass recommended to use while the drilling is tacking place

2.5 Board Cutout

this is the last operation for the PCB change the bit to the Flat end milling bit load the g-code for the board cutout move the spindle to 0,0,0 press start use the vacuum to remove the fiberglass recommended to use while the milling is tacking place

2.6 2-side PCB

load the alignment drill g-code change the bit to your drill bit use the x and y movement controls to move the spindle to x = 0 and y = 0 and then use the z axis control to move the drill bit down to the copper and set z = 0 press the start button

declamp the laminate flip the laminate to the other side use the alignment holes to align the laminate clamp the laminate down to the work bed load the bottom isolation routing g-code change to the spiral bit move the spindle to 0,0,0 and press auto level press start button

3 Links:

3.1 flatcam:

http://flatcam.org/manual/index.html

3.2 wegstr manual pdf:

https://testglassware.com/wp-content/uploads/2022/05/MANUAL-ITEM-1.pdf